Process Characterization Plan

A-Mab Drug Substance — Protein A Chromatography (Step 5) — PCP-005

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Field	Value
Document ID	PCP-005
Document class	Process Characterization Plan
Title	Process Characterization Plan: Protein A Chromatography (Step 5)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ANOVA, analysis of variance · CCD, central-composite design · CEX, cation-exchange chromatography · CPP, critical process parameter · CQA, critical quality attribute · CV, column volume · DoE, design of experiments · DS, drug substance · ELISA, enzyme-linked immunosorbent assay · GPP, general process parameter · HCP, host-cell protein · ICH, International Council for Harmonisation · KPP, key process parameter · MSAT, Manufacturing Science and Technology · NOR, normal operating range · PAR, proven acceptable range · PPQ, process performance qualification · QbD, Quality by Design · RRF, risk ranking and filtering · RSM, response-surface methodology · SDM, scale-down model · SEC, size-exclusion chromatography · WC-CPP, well-controlled CPP.

1 Purpose and scope

This Process Characterization Plan (**PCP-005**) defines the Stage 1 (Process Design) characterization of the A-Mab Protein A capture chromatography step (Step 5), the affinity-capture operation that binds and concentrates the product from the clarified harvest and provides the first and principal reduction of host-cell protein (HCP) and residual DNA. The plan specifies the objectives, the attributes in scope, the process parameters and ranges to be characterized, the experimental designs, the analytical methods, the sampling plan, the statistical-analysis approach, and the acceptance and decision criteria by which each parameter will be classified and a multivariate operating region established.

Scope. The commercial-scale Protein A capture step, characterized on a qualified scale-down chromatography column. The upstream harvest (Step 4) and the downstream polishing steps are addressed only where they define the feed to, or clear the impurity load from, this step; their characterization is out of scope. This plan governs execution only — the results are reported in the paired Process Characterization Report (**PCR-005**).

Basis. The unit-operation model, quality-attribute framework and risk methodology follow the A-Mab case study (CMC Biotech Working Group 2009); the Stage 1 approach and report structure follow PDA TR 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023); the QbD framework follows ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012) and the lifecycle approach of the FDA 2011 guidance (U.S. Food and Drug Administration 2011).

2 Objectives

The characterization will:

1. establish the quantitative relationships between the Protein A process parameters and the eluate-pool HCP, the step yield and the leached Protein A;
2. identify the significant main effects, two-factor interactions and curvature;
3. define the multivariate operating region over which the eluate-pool HCP presented to the polishing steps is controlled;
4. establish proven acceptable ranges (PARs) and confirm normal operating ranges (NORs) for each parameter;
5. classify each parameter for its risk of impact to CQAs and to process performance (CPP / WC-CPP / KPP / GPP); and
6. provide the process understanding and ranges that justify the Stage 2 operating conditions and the drug-substance control strategy.

3 Related documents

This plan is executed under the master documents below and is paired with its report; the full package is cross-referenced for traceability.

Table 3.1: Related documents in the A-Mab process-characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-005	Process Characterization Report (Protein A Chromatography (Step 5))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The governing procedures and validated analytical methods referenced by this plan are listed in Table 3.2 (numbers are placeholders in this synthetic set).

Table 3.2: Referenced standard operating procedures and analytical method validations.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2007	Operation of the Protein A Capture Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

Protein A chromatography is the capture step of the A-Mab purification train: the clarified harvest is loaded onto a Protein A affinity resin that binds the Fc region of the antibody, contaminants are removed in the wash, and the product is recovered by a low-pH elution (CMC Biotech Working Group 2009). It does not form or modify the product-quality CQAs established in cell culture; its role is to capture and concentrate the product, to provide the first and largest reduction of HCP and residual DNA, and to introduce a single process-related impurity — leached Protein A — that is subsequently cleared. Platform knowledge established that the eluate HCP burden increases with column load and with lower elution pH, that the two covary through a load × elution-pH interaction (A-Mab Figure 4.2), that recovery is governed by the load flow rate and the extent of the elution pool collected, and that a small, characteristic amount of Protein A leaches under low-pH elution. This knowledge frames the attributes in scope and the prioritization of parameters for characterization.

4.2 Quality attributes in scope

Protein A introduces one CQA — leached Protein A (Table 4.1) — and is the principal clearance step for host-cell protein and residual DNA, which are formed upstream and specified at the drug substance. Criticality is scored on the A-Mab continuum using Tool #1 (Risk Score = Impact × Uncertainty); attributes rated High (H) or Very High (VH) are treated as *Critical* (CMC Biotech Working Group 2009; International Council for Harmonisation 2023).

Table 4.1: CQA introduced by the Protein A step, with acceptance criterion and criticality (Tool #1 = Impact × Uncertainty).

CQA	Category	Acceptance	Criticality	Tool #1
Leached Protein A	process impurity	0–5 ppm	L-M	16

The eluate-pool HCP is not a released specification but an in-process attribute whose control determines the impurity load presented to the polishing steps and the margin to the 100 ng/mg drug-substance HCP limit; it is the primary response of the study.

4.3 Risk-based prioritization of parameters

Consistent with the A-Mab lifecycle of iterative risk assessments and the risk flow of PDA TR 60, the study scope is set by the Pre-Characterization Process Risk Assessment (**RA-001**). A risk-ranking-and-filtering (RRF) screen scores each parameter for its potential main-effect and interaction impact and assigns the study type accordingly: protein load and elution pH (credible impact to the eluate-pool HCP) and the load flow rate and end-of-pool-collect (recovery) are studied in the multivariate DoE; operating temperature and bed height, expected to affect only process performance over a wide range, are studied univariately (CMC Biotech Working Group 2009). The resulting parameter set and study-type assignment are given in Table 6.1.

5 Materials and methods

5.1 Scale-down model and its qualification

Characterization will be performed on a scale-down Protein A column operated at the commercial-process equivalents for resin, bed height, residence time, buffer system, load ratio and elution strategy (**SOP-2007**, **SOP-2008**). Prior to characterization the SDM will be qualified against at-scale reference data (**SOP-1001**) by comparing the operational parameters and, critically, the input and output attributes — step yield, eluate-pool HCP and residual DNA, and leached Protein A — to confirm the model is representative and suitable to support operating-range and PAR claims (Parenteral Drug Association 2013). The centre-point replicates of each design provide an in-study estimate of model reproducibility (pure error) used in the lack-of-fit assessment.

5.2 Operation

The clarified harvest is loaded onto the equilibrated Protein A column to the target load ratio; the column is washed to remove weakly bound host-cell protein, DNA and media components; and the product is recovered by a low-pH elution, with the eluate pool collected between defined start and end criteria (the end-of-pool-collect, expressed in column volumes). Buffers, resin lot and column packing are controlled to platform specifications (**SOP-2008**) and are treated as identity/quality-controlled inputs rather than characterized variables (CMC Biotech Working Group 2009).

5.3 Analytical methods

The step responses will be measured by the validated methods in Table 5.1; method performance characteristics are per the cited validation reports.

Table 5.1: Validated analytical methods for the characterization.

Reference	Title	Type
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

5.4 Sampling plan

Each DoE run will be sampled at the eluate pool for HCP, residual DNA and leached Protein A, and the load and pool product mass will be recorded for the step-yield determination; in-process readings (pH, conductivity, UV) will confirm the chromatographic profile. Centre-point runs are replicated within each design (three in screening, four in the response-surface design) to quantify reproducibility and pure error.

5.5 Statistical methods

Designs will be constructed and analyzed in coded factors, with each parameter's characterization range mapped so that the set-point is 0 and the range edges are ±1 (**SOP-1002**). Screening data will be analyzed by ordinary-least-squares regression of each response on the main effects and all two-factor interactions; factor effects are reported as twice the coded coefficient. Response-surface data will be analyzed with the full quadratic model (main effects, two-factor interactions and pure-quadratic terms). Significance will be assessed at = 0.05. Model adequacy will be judged by R^2, adjusted R^2, predicted R^2 (from the PRESS statistic), the overall model F-test, and an ANOVA lack-of-fit test against the centre-point pure error; a response with no significant parameter effect will be reported as robust rather than modelled predictively. The operating region will be defined as the multivariate region of protein load and elution pH over which the eluate-pool HCP is controlled, and commercial-scale capability of the drug-substance impurity CQAs governed by the step will be estimated by Monte-Carlo simulation with each parameter varied within its NOR.

6 Study design

6.1 Factors, ranges and study type

Six parameters are in scope (Table 6.1). The characterization range of each factor ("Range studied") defines the edges of the explored knowledge space and the basis of the PAR; the NOR is the tighter routine-control range. Classification is an output of the study and is reported in PCR-005. The DoE factor coding for the four multivariate factors is given in Table 6.2.

Table 6.1: Protein A parameters, set-points, characterization ranges and planned study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Protein load	g/L resin	30	10–50	20–40	multivariate
Elution buffer pH	pH	3.55	3.2–3.9	3.4–3.7	multivariate
Load flow rate	cm/hr	200	100–300	150–250	multivariate
End of pool collect	CV	2.6	2–3.2	2.3–2.9	multivariate
Operating temperature	°C	22	15–30	18–26	univariate
Bed height	cm	20	10–30	15–25	univariate

Table 6.2: Design-of-experiments factor legend (coded A–D).

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Elution buffer pH	pH	screening + RSM
C	Load flow rate	cm/hr	screening + RSM
D	End of pool collect	CV	screening + RSM

6.2 Screening design

Factor screening will use a two-level full factorial in the four multivariate factors (A–D) with three centre-point replicates, 19 runs in total. The full factorial estimates all four main effects and all six two-factor interactions without confounding. Three responses will be recorded for each run (eluate-pool HCP, step yield and leached Protein A). The planned coded design is given in Appendix A.

6.3 Response-surface design

The four multivariate factors will be carried into a face-centred central-composite design with four centre-point replicates, 28 runs in total, enabling estimation of curvature in addition to main effects and interactions. The planned coded design is given in Appendix B.

6.4 Univariate assessment

Operating temperature and bed height will be evaluated one factor at a time across their characterization ranges for their effect on step yield, eluate-pool HCP and the chromatographic profile.

7 Acceptance and decision criteria

The study is acceptable when:

1. the SDM qualification confirms representativeness of commercial scale (**SOP-1001**);
2. the fitted response-surface models for the eluate-pool HCP and the step yield are adequate — target R^2 0.90, adjusted R^2 0.80, no significant lack of fit ($p > 0.05$ against the centre-point pure error), and residual diagnostics without systematic structure (a response with no significant parameter effect is reported as robust rather than modelled); and
3. a multivariate operating region exists over which the eluate-pool HCP is controlled at the set-point and across each parameter's NOR.

Each parameter will then be classified per the A-Mab continuum (CMC Biotech Working Group 2009):

- a parameter whose variability significantly impacts a CQA is a **CPP**; where the impact is reliably controlled within the operating region (low risk of leaving it) the parameter is a **well-controlled CPP (WC-CPP)**;
- a parameter that does not significantly impact a CQA but affects process performance or consistency is a **KPP**; otherwise it is a **GPP**;
- consistent with the post-characterization FMEA convention, a parameter that affects a CQA with Severity   8, or that carries an initial (pre-mitigation) risk priority number > 72, is designated a CPP before mitigation by the control strategy.

The operating region, classifications and PARs established here will justify the Stage 2 operating ranges and feed the drug-substance control strategy.

8 Data management and integrity

All raw data, DoE designs, analytical results and statistical analyses will be recorded and retained under the site data-integrity procedures (ALCOA+). Analyses will be performed with version-controlled scripts and reviewed by a second analyst; designs, models and the derived operating region will be archived with the report. Any deviation from this plan will be documented and assessed for impact in PCR-005.

9 Roles and responsibilities

Table 9.1: Roles and responsibilities.

Function	Responsibility
Process Development / MSAT	Author the plan; design, execute and analyze the DoE; author PCR-005
Analytical Development	Perform released and in-process testing under the referenced AMV methods
Manufacturing Science	Qualify the scale-down model; confirm commercial-scale representativeness
Statistics	Review the design, the statistical analysis and the operating-region definition
Quality Assurance	Approve the plan and the report; confirm parameter classifications

10 Deliverables and schedule

The deliverable of this plan is the Process Characterization Report (**PCR-005**), reporting the executed designs, the significant effects, the operating region, the parameter classification and the impurity-CQA outcomes. PCR-005 rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment. Execution follows SDM qualification and precedes the Stage 2 PPQ campaign.

11 Risks and assumptions

The plan assumes the SDM is representative of commercial scale (confirmed by qualification) and that the platform analytical methods are validated and in control. The principal risk is that the absolute eluate-pool HCP or the resin-age contribution to leached Protein A shift at commercial scale or over the resin life-cycle; this is mitigated by defining the operating region on the qualified SDM, confirming it at scale in Stage 2, and carrying the WC-CPP ranges forward with in-process monitoring and resin life-cycle control.

12 Approvals

Approval of this plan authorizes execution of the characterization described herein. Signatures are recorded in Table 1.

References

CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.

International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.

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International Society for Pharmaceutical Engineering. 2023. *Good Practice Guide: Practical Implementation of the Lifecycle Approach to Process Validation*. ISPE.

Parenteral Drug Association. 2013. *Technical Report No. 60: Process Validation — a Lifecycle Approach*. PDA.

U.S. Food and Drug Administration. 2011. *Guidance for Industry: Process Validation — General Principles and Practices*. FDA.

Appendix A — Planned screening design

Coded two-level full-factorial design (factors A–D per Table 6.2; coded levels $-1/0/+1$). Responses will be recorded at execution and reported in PCR-005.

Table 12.1: Appendix A — planned screening design (coded factor settings).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

Appendix B — Planned response-surface design

Coded face-centred central-composite design (factors A–D; coded levels $-1/0/+1$).

Table 12.2: Appendix B — planned response-surface design (coded factor settings).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0